# Experiment 17: Write a Prolog Program to Sum the Integers from 1 to N

**Aim:** To implement a Prolog program to calculate the sum of integers from 1 to N.

## Procedure:

1. Open a Prolog programming environment.
2. Define the base case for the sum.
3. Define the recursive rule.
4. Accept the value of N through a query.
5. Calculate the sum recursively.
6. Display the final sum.

AI_exp17.pl

File Edit View

```
sum(0, 0).
sum(N, S) :-
    N > 0,
    N1 is N - 1,
    sum(N1, S1),
    S is N + S1.
```

## Output:

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```
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

3 ?- consult('C:/Users/acer/Downloads/AI_exp17.pl').
true.

4 ?- sum(5, S).
S = 15
```

**Result:** Thus, a Prolog program to calculate the sum of integers from 1 to N was implemented successfully.